

Benedict Leung

HUMAN – COMPUTER INTERACTION RESEARCHER

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Human–computer interaction researcher focused on human–AI interaction and novel input modalities, such as pen and stylus, eye-tracking, and brain–computer interfaces. I take interactive systems from concept to working prototype to controlled user study, with publications at top-tier venues including CHI and UIST. Currently working on human–AI collaboration with large language models.

EDUCATION

Computer Science (MSc)

09 / 2023 – 08 / 2025

Ontario Tech University

- Led thesis research on generative-AI–assisted pen annotation of documents, published at CHI 2026.
- **Advisors:** Mariana Shimabukuro and Christopher Collins

Computer Science (BSc Hons), Minor in Mathematics

09 / 2017 – 05 / 2022

Ontario Tech University

- Led thesis research reinventing hands-free camera interaction, published at UIST 2024.
- **Advisor:** Christopher Collins
- President’s List (Fall 2020 – Winter 2022); Dean’s List (Fall 2017).

EXPERIENCE

Research Associate

09 / 2025 – present

Ontario Tech University — Vialab

- Built a French-vocabulary learning app in which users hunt for and identify real-world objects to learn words in context.
- Contributing to a gaze-aware interaction study (ETRA 2026) that detects distraction and guides attention recovery during foreign-language video learning.

Undergraduate Student Research Fellow

09 / 2022 – 08 / 2023

Ontario Tech University — Vialab

- Led research on gaze-driven question generation for personalized learning from short educational videos.
- Ran user studies and published the system as GazeQ-GPT (Graphics Interface 2025).

Undergraduate Student Researcher

05 / 2022 – 08 / 2022

Ontario Tech University — Vialab

- Led research on crossing-based pen-selection techniques for stylus and touch interfaces.
- Designed and evaluated the techniques through a controlled user study.

Undergraduate Thesis

09 / 2021 – 04 / 2022

Ontario Tech University — Vialab

- Designed and built a touchless camera controlled through mental commands and hand gestures.
- Combined a brain–computer interface with eye-tracking glasses for hands-free control; published as NeuroSight (UIST 2024).

Research Assistant

07 / 2021 – 12 / 2021

Ontario Tech University

- Built and deployed a web-based breadboard circuit simulator used to teach digital logic design.

Teaching Assistant

09 / 2022 – 05 / 2025

Ontario Tech University

- Programming Workshop I (Fall 2022) · Programming Workshop II (Winter 2023)
- Software Systems Dev. & Integ. (Winter 2024, 2025) · Human Computer Interaction (Fall 2023, 2024)

TECHNICAL SKILLS

Languages & Tools	JavaScript / TypeScript, HTML / CSS, Python, React, Git, R, C / C++, SQL, PyTorch
HCI & UX Research	Controlled experiments, usability studies, mixed-methods, study design, statistical analysis, prototyping
Interaction & Sensing	Eye-tracking, brain-computer interfaces, IMU sensors, pen / stylus input, gaze-based interaction
AI & LLMs	LLM application development, prompt engineering, human-AI interaction design

PUBLICATIONS

Note: In HCI, peer-reviewed conferences are the primary archival venue, and top conferences like CHI and UIST are highly selective with average acceptance rates ~25%.

- [C4] Mohammed Ahmed, **Benedict Leung**, Mariana Shimabukuro, Christopher Collins. Don't Wanna Miss a Thing: Gaze-Aware Implicit Interventions for Distraction Recovery in Foreign-Language Videos. **ETRA 2026**
- [C3] **Benedict Leung**, Mariana Shimabukuro, Christopher Collins. AnnotateGPT: Designing Human-AI Collaboration in Pen-Based Document Annotation. **CHI 2026**
- [C2] Neel Shah, **Benedict Leung**, Mariana Shimabukuro, Ali Neshati. SwipeSense: Exploring the Feasibility of Back-of-Device Swipe Interaction Using Built-In IMU Sensors. **MobileHCI 2025**
- [C1] **Benedict Leung**, Mariana Shimabukuro, Matthew Chan, Christopher Collins. GazeQ-GPT: Gaze-Driven Question Generation for Personalized Learning from Short Educational Videos. **Graphics Interface 2025**
- [EA1] **Benedict Leung**, Mariana Shimabukuro, Christopher Collins. NeuroSight: Combining Eye-Tracking and Brain-Computer Interfaces for Context-Aware Hand-Free Camera Interaction. **UIST EA 2024**

THESES

[T2] MSc Thesis · Implicit Pen Annotation Assisted by Large Language Models

Ontario Tech University

- Published to **CHI 2026**. Full thesis available [here](#).

[T1] BSc Thesis · Touch Free Camera: Mental Commands and Hand Gestures

Ontario Tech University

- Published to **UIST EA 2024**. Full thesis available [here](#).

HONOURS & AWARDS

NSERC USRA — \$8,400

05 / 2022

Ontario Tech University — Vialab

- Competitive national award for 16 weeks of full-time undergraduate research, granted on academic merit and funded by NSERC.

In-Course Scholarship

09 / 2021 – 09 / 2022

Ontario Tech University

- Awarded for maintaining a 4.0–4.3 GPA over the academic year.